AdopTree

Cecilie Städe Kristensen
Aalborg University
Aalborg, Denmark
Cskr20@student.aau.dk

ABSTRACT
Urban trees play an important role in improving city environments, from absorbing CO_2 to supporting mental health. Even after all that, the same tress are neglected because of a lack of awareness and care. Research show how people are willing to help care for urban trees if they are given the right tools and instructions. In this paper, I will present how I used the five stages of the Design Thinking Process to develop a high-fidelity prototype for AdopTree. AdopTree is an app that encourages users to adopt urban trees nearby, receive AI-generated health diagnosis through check-ins, and get simple and clear instructions and tools to help users care for their trees. The app encourages a meaningful and consistent engagement from the users. This paper will be presenting my research and ideation process behind AdopTree, as well as a plan for a usability test of my prototype. The paper concludes with a discussion on limitations and possible ideas for future iterations.

Keywords
Urban cree care, AI based check-ins, Tree adoption, location-based map.

INTRODUCTION
Somewhere in your city, maybe even in your backyard, a tree is suffering in silence. All alone, withering and neglected by the very same people it helps out. That same tree, once upon a time, stood tall, leafy, and full of potential. But now? It is silently screaming for help, desperate for a check-in and someone to care for it. While we doom-scroll climate news and debate whether to buy oat milk or almond milk. The trees are cleaning our air, cooling our cities, and absorbing the CO_2 we keep throwing into the atmosphere [4]. And what do they get in return? No one to help them in their battle against pollution, vandalism, neglect, and the occasional weed whacker gone rogue. But now it's your chance to give back and become a tree parent, with *AdopTree*.

With *AdopTree*, you now have the opportunity to adopt a tree near your and take part in its care and well-being. Regardless of if you are a newly baked tree parent, who just want something low maintenance, or if you are a green thumb veteran, ready to go fingers first into the war against urban tree neglect. There is a branch with your name on it. If you are an inexperienced tree parent, don't worry, start with a healthy tree, and help prevent future distress. Or, if you are in the marked for a tree with less baggage, and no previous bad influences, you can adopt a newly planted tree. But if you are feeling bold, adopt a tree that is already in need of help, and save a live. All you have to do, is find and adopt a tree near you, check-in on your tree baby by taking photos, and get the app's AI scanners diagnosis, along with the tools to help your tree live its best life.

The urban trees suffer because of a lack of awareness, as well as a system of care [7]. Even though trees appear self-sufficient, because they can grow without human involvement from the get go, they are still vulnerable and in need of a lot more care than what it looks like [13]. Without support, the trees struggle to handle all the challenges they face day to day, especially in the city. Research shows that, a large number of citizens wants to help and be engaged in caring for nature, if they just have the right tools [2].

These reflections on the neglect of urban trees and people's lack of awareness of the nature around them, laid the ground for this design project. This is supported by related works and insights from user research, exploring how care and responsibility for urban trees, can be promoted through technology. Followed by an ideation process, that involves brainstorming, affinity diagramming, exploratory sketching, and a high-fidelity prototype. Lastly, the paper shows a plan for evaluating the prototype through a usability test, and a discussion on limitations, and possible ideas for future iterations.

METHOD
I chose the Design Thinking Process to approach the problem of urban tree neglect. The Design Thinking Process is a non-linear, iterative process which consists

of five stages: *Empathize, Define, Ideate, Prototype,* and *Test* [16].

During the Empathize stage, I tried to understand the problem from multiple perspectives, and set my initial assumptions aside [16]. I gathered and reviewed literature on urban trees including their ecological needs, the challenges of maintaining them, and the lack of involvement in their care. Besides that, I conducted unstructured interviews with three people living different lifestyles. This was done to get different insights into awareness of urban tree neglect and get possible motivations to get engagement.

In the Define stage, I analyzed findings from the literature, and the answers from the interviews, to identify problems and see possible needs. At the end of the Define stage, I was able to formulate a problem statement. At that point, I was able to develop different user requirements, which served as the base for the following design stages [17].

In the Ideate stage, I explored different solutions to the problem through brainstorming, affinity diagramming and exploratory sketching. This phase resulted in two main concepts: gamified tree challenges and tree adoption with AI check-ins. After that, I evaluated both concepts based on achievability, motivation for engagement, and the ecological impact. In the prototype stage, I developed a high-fidelity prototype of the chosen concept using Figma [6]. The prototype was based on the previously defined user requirements, which became key features of my prototype, *AdopTree*. At the end, I made a plan for a usability test, which would make it possible to evaluate the prototype.

Because Design Thinking is an iterative process, some of the stages were revisited throughout my design process [16]. For example, while sketching and designing the high-fidelity prototype, I returned to both the literature and the interviews to get more inspiration.

RELATED WORK

During the design process, I looked into literature and articles related to urban tree care, ecology, and public engagement in environmental care. I also looked into apps related to those topics, to see how preexisting technology has been used to support urban tree care. The goal was to get insights into how interaction and awareness about urban trees, could be encouraged through design.

A report from the University of Copenhagen's Videntjeneste, describes how the urban trees require a lot more care than we often assume. This is primarily due to their restricted growing conditions, air pollution, and human interference [4]. The article points out, even though trees are often perceived as independent, they are actually very dependent on external maintenance and monitoring. To be able to survive in urban settings. There are similar problematics in an DR article, on how municipalities are struggling to care for urban trees. Mostly because, they are not able to detect the early signs of diseases or damage on the trees, without dedicated systems. This causes municipalities to have a larger loss of urban trees than necessary [3]. These views are supported by an article from Woodland Trust. The article highlights how different three species improve our quality of life in so many aspects. They help with everything from, absorbing excess carbon dioxide (CO2), and help combat climate change, to improving both our physical and mental health. This just shows how fragile these benefits are, without external care and maintenance [19].

The literature shows that, while urban trees do very important work for the environment, they are often neglected. Due to a lack of awareness, responsibility for the trees, and accessible tools. But studies and articles, show that people are open to help protect nature, they just need clear and meaningful ways to do it [10].

I also looked into how technology can fill this gap and get people more involved in caring for urban trees. This was done by looking into preexisting apps, that aims to raise awareness about urban green spaces. Healthy Trees, Healthy Cities (HTHC) [21] is a mobile app, developed by The Nature Conservancy and the U.S. Forest Service. It is designed to help users maintain and monitor healthy trees and promote the long-term health of urban trees, by providing free tools and resources [8]. HTHC is offering the users modules, such as tree planting, watering, trimming, monitoring tree health, and noticing pests and diseases. HTHC gives users with no previous knowledge, the chance to help out by sharing useful information, which municipalities and researchers can use [21]. The simple design of the app, and a provided step-by-step guide is available, to make it easier for the user to get started [21].

Another app I came across was Pl@ntNet [11]. It is a citizen science platform that uses artificial intelligence (AI) to help users identify, tens of thousands of plant species, just by photographing them. The app lets users create an account where they can share their observations, which can be reviewed by the established community in the app. The users have the opportunity to confirm the name of a plant species, as well as the opportunity to suggest another observation. This can only be done if they have preexisting knowledge about plants. Only the observations that are confirmed with enough confidence, get added to the apps public database and is then used to train the AI. The app encourages the users to share their GPS locations, with their submitted photos. This feature makes it possible to map species geographically, and by that provide valuable data for researchers and promote environmental monitoring [22]. The app helps users learn about nature by giving them the ability to identify plants through photos. It shows how a simple design, combined with technology can promote engagement with nature.

These findings show that digital solutions can help people take care of nature, by making it easier for them to learn, take action, and be able to see the results of their efforts. This fits with the research, about what gets people involved, such as seeing their impact physically.

User research

To gain further insight on user experiences, their thoughts on urban trees, awareness gabs, and what might motivate them to take the necessary steps to care for them. I had three unstructured interviews, with individuals with different lifestyles. The reasoning behind the chosen participants, was to get a broader perspective. The unstructured interview made it possible to have a natural flow in the conversation. They could talk freely about their thoughts and experiences, while giving me the chance to ask questions and dig deeper into specific parts of the conversations. This helped me gather detailed information and observe a pattern in the responses [18].

I started each interview with the same question:

Do you notice the trees around you, and do you ever wonder about the state of their health?

This was done to start out, seeing the differences of awareness from the participants. There was a theme amongst the answer from the participants. They liked the aesthetics of the trees, but they didn't really think deeper about it, as to who takes care of them. One of the participants said:

I never really thought about it, I always kind of thought the city had someone look at them (P2)

This answer is an example of a recurring theme of passive appreciation, with a lack of awareness and responsibility. To the question, about the idea of helping take care of urban trees, through an app. All three participants were open to the concept, but with conditions. Participant 1 had the condition:

I have a pretty busy lifestyle, so I would be more open to the idea if the tree was near me like in walking distance so I could do it whenever I wanted

Where participant 3 had the condition:

I don't really know anything about how to help a plant or tree, so I would need specific directions on what to do

These conditions formed two of the key requirements to my concepts. There has to be a map feature or a way to put in your location and there needs to be clear instructions. Another answer that became a key feature, was about how it would feel rewarding to get feedback or a need for motivation, like the feeling of being needed. This answer showed how important it could be to get feedback or visually see the progress of the users' interactions. In the end this answer helped make the decision on the which final concept to choose. The interviews made it clear, that there is a pattern of unawareness of the neglect and the challenges that urban trees face. It also showed the positives, of people wanting to help as long as they have the tools and clear directions.

The problem domain

Based on the insights I gathered from my interviews and the literature, I formulated the following problem statement: "*How can I encourage people to engage in the care of urban trees, in an accessible and motivating way through technology?*"

DESIGNING ADOPTREE

I started my ideation process with a focus on urban trees. This decision was influenced both by answers from the interviews and the literature. During one of the interviews, the participant talked about the need for convenience on busy days. So they would be more likely to engage and care for a tree if it was in a walking distance. This fit into the findings from my research, which addressed how urban trees are particularly vulnerable, and need care and monitoring [3]. With this focus in mind, I started my ideation process with a brainstorming session to generate early design ideas. During my brainstorming session, I wrote down different ideas and concepts which related to tree care and user engagement, with technology involved. After that I used affinity diagramming, which is used to group related ideas into clusters [1]. By using this method, I was able to see underlying patterns and mix different rough ideas to develop two main concepts:

Concept 1: Gamified tree challenges.

Concept 2: Tree adoption with AI check-ins.

This was done in Miro [14] using sticky notes, to make the process of moving and organizing the ideas easier (Figure 1).

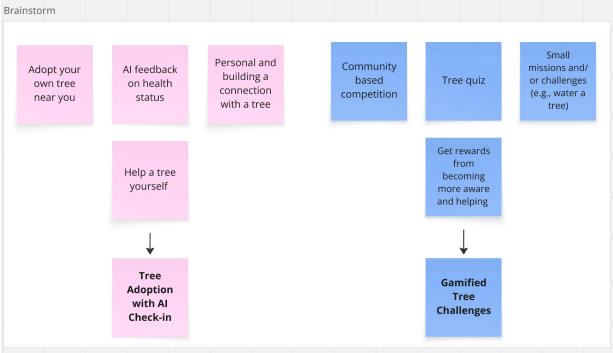

Figure 1: Brainstorming ideas

To further understand and compare the final concepts, I made low-fidelity sketches for both concepts. By using exploratory sketching, I was able to visualize my ideas, main features and interactions [20]. By implementing this method, I was able to see some of the strengths and the limitations for each concept, while drawing the sketches. Each of the sketches contains arrows and notes to explain some of the important features and flows. This helped me choose which concept was the best to work further on and get an understanding of how it would work in real life [20].

Concept 1: Gamified Tree Challenges

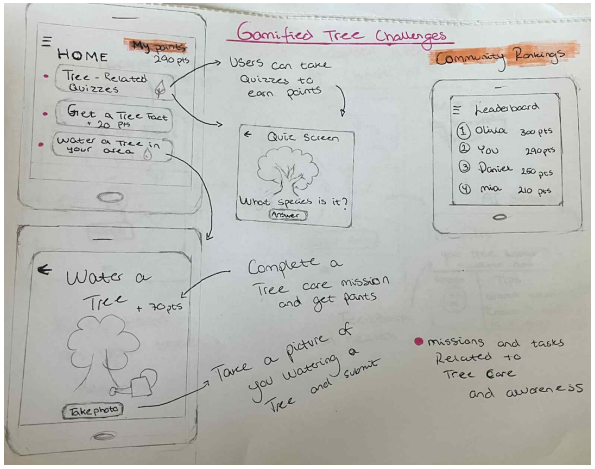

Figure 2 Sketch of Gamified Tree Challenges

Concept 1 (Figure 2) is an app, with the intention of motivating users to engage with urban tree care, through playful and rewarding activities. The user can choose between tasks and missions (e.g., watering a tree or learning a fact). This encourages the user to become more aware of the trees around them. The app incorporates gamification and encourages users to interact more with tree health. This is done by making it possible for the user to receive points and compare points with other users. This concept is more community based, with competitive elements. It encourages users to take small but still meaningful actions, that supports urban tree health, as well as raises awareness about urban tree needs.

Concept 2: Tree Adoption with AI Check-ins

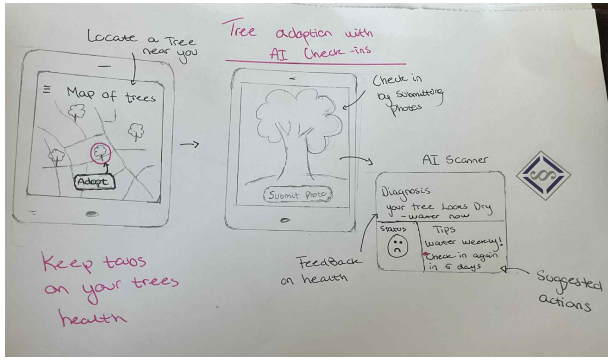

Figure 3 Sketch of Tree Adoption with AI Check-in's

Concept 2 (Figure 3) is an app, that aims to create a personal and hopefully long-term connection between the user and their adopted tree. By giving the user a responsibility for their selected tree. The user is encouraged to make regular check-ins, and get AI generated feedback. This helps them getting the tools to care for their tree. The user can adopt an available tree near them, from a tree map. They can also keep tabs on their tree, by submitting photos and getting health statuses. This concept is more personal and aims to give the user a sense of ownership and therefore make the care more consistent and properly.

Selection process

To help me decide which of the two concepts to choose, I evaluated both ideas using the following criteria.

Achievability: if the concept could be done, with the technologies available today?

Motivation for engagement: if the users would feel encouraged to take care of urban trees?

The ecological impact: if the final concept would contribute in a meaningful way, in terms of the well-being of the urban trees?

I ended up choosing the second concept, about tree adoption with AI check-ins. Even though the gamified tree challenge concept had its potentials. Seen from a larger perspective, it was too superficial and less focused on the actual care for urban trees. Where the second concept gives the users the opportunity to create an emotional connection. Which promotes long-term care for the adopted trees. The second concept was also both supported by the research and the interviews. One of the things the research showed, was how urban trees require a lot more care than we often assume. It also revealed that people are willing to help care for them, they just need clear and meaningful ways to do it. This aligns with the key features of the second concept and how the AI check-ins give the tools to do so. One of the participants from the interview, pointed out how it was important for them to see their impact, and they would be more consistent if it was in a walking distance. These needs are met with the AI feedback, and the location-based map.

Prototype of *AdopTree*

After choosing my final concept, I named it *AdopTree* and began developing a high-fidelity prototype. Choosing to create a high-fidelity prototype, made it possible to see what the idea would look like. Further fix or resolve the functions that aren't working, and gather future feedback before making the actual product [12]. My prototype is made as a high-fidelity prototype, using Figma [6]. It gives an idea of what the flow would look like, and how the app would function [9]. The prototype consists of 9 frames, where each of the frames will be referred to with a specific letter to prevent confusion. I was able to make a set of requirements, based on the answers from the interviews and the literature. These served as the base for creating *AdopTree* and will be pointed out later in this section.

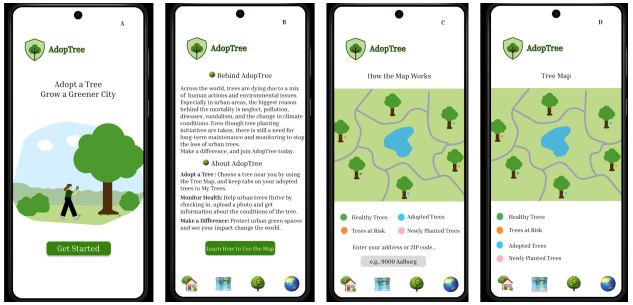

Figure 4 A) Welcome page B) Homepage C) Introduction to the Tree Map D) The Tree Map

While designing the interface for the prototype, I had to choose which colors to use for my design. I ended up using a variety of green colors, to represent the nature theme. Green is a very calming color and easy on the eyes [5]. It became a recurring theme of using natural and calming colors, but with exceptions when it came to features that had to stand out.

I started making a *Welcome page* (A). This includes an image and the tagline *"Adopt a Tree, Grow a Greener City"* to shows the apps intentions. At the bottom of the page there is a button labeled "*Get Started*", to get the user to explore the app further. After the user presses the button, they are sent to the apps *Homepage* (B). This page is an introduction to the concept behind the app. It goes into detail about the neglect that urban trees face and the apps mission. After that, the user is met with clear and simple instructions, to provide some knowledge about the features. At the bottom of the page, the user is met with a *"Learn How to Use the Map"* button, which sends them to the *How the Map Works* page (C). This page is an introduction and guide, as to how the map works. There is a set of colored dots, to show the trees status: *At Risk*, *Healthy*, *Newly Planted*, and *Already Adopted*. This was designed make things simpler and help the user see which trees are available, so they can choose one that suits them and feel manageable. This page includes a search bar where the user can enter their ZIP code or address, to find trees near them. This function was a requirement, so the app would be more convenient for the user. When the user presses enter after writing their ZIP code or address, they are sent to the actual map, called *Tree Map* (D). Here, the user can see the urban trees in their area, every tree has a little dot that indicates the trees status. The user can click on the trees and is sent to a new page for that specific tree (E). Then the user is met with information about the tree's species (e.g., Oak Tree), a photo of the actual tree, its location and status. These features aim to encourage the user to feel a personal connection and responsibility, like it's their tree, a real tree waiting for someone to care for it. This is especially important for the trees marked as "*At Risk*", as it aims to create a feeling of urgency and empathy, that hopefully turns awareness into action. At the bottom of the page, there is an "*Adopt This Tree*" button, to encourage the user to adopt the tree they are looking at. If the tree is already adopted, there is no adopt button.

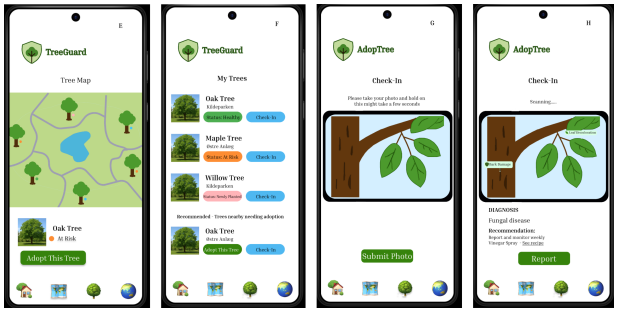

Figure 5 E) Selected Tree from map (At Risk) F) My Trees page G) Check-in H) AI Health generator

At the bottom of the app there is a navigating menu, with short cuts to the *Homepage* (B), the *Tree Map* (D), the *My Trees* (F) and the *Our Impact* (L) pages. When the user is on one of the pages, the icon has a little shadow underneath it, to make clear where they are on the app.

The "*My Trees*" page (F) is created to give the user a view of their adopted trees, and encourage ongoing care. For each of the adopted trees, they have their species, location and photo implemented. This helps the user differentiate between their adopted trees and locate then. This is also to keep the user accountable for their adopted trees, by showing their statuses. This is also a reminder, that they have taken responsibility for those trees. At the bottom of the page there is a recommendation of trees nearby, that needs adoption. This is to remind the users that more trees need care and attention, without being too intrusive. Next to the tree's status, there is a blue check-in button that directs the user to the check-in page for that specific tree (G-H).

One of the main features to be implemented in the future, is the AI health scanner. This should allow the user to make visual check-ins on their adopted trees. This is to promote user engagement and meet the key requirements. Like the need for clear instructions and the tools to give the needed care. The user is instructed to take a photo of their adopted tree and submit that photo with a "*Submit Photo"* button (G). Because this is still a prototype, there is no backend processing. The intention is to use image recognition, to analyze the users' submitted photos. After the photo is submitted, the AI scans the photo of the tree for any visual indicators of distress or diseases (e.g., leaf discoloration, bark damage). When the scan is done, the app gives the user a diagnosis (e.g., fungal disease), and the recommended care (H). This feature aims to make knowledge about urban tree care more accessible and give a simple step-by-step guide. The goal is to give the user a sense of ownership and accountability, by showing visual evidence of tree neglect. At the bottom of the page there is a "*Report*" button, which in the future could be used to save the data and give the user the ability to have a History of Health page, so they can follow their trees progress and development through time.

Figure 6 L) Our Impact page

The last feature of the navigation menu is the globe, which takes the user to the *"Our Impact"* page (L). This page is created, so that the users can see their shared progress and the community growing. The page shows a visual representation of all the trees being cared for across the world. Each of the newly adopted trees appears as a glowing point on the globe. At the bottom of the page there is a counter, which in further works would be a real-time counter, that shows the total number of check-ins and how many new members has joined the community, each week.

EVALUATION

To evaluate *AdopTree*, I would do a usability test using a task-based scenario approach. This evaluation method gives a clear visual of what to improve, when it comes to an interactive design [15]. I would choose this method, because my prototype has interactive features, like tree adoption, map navigation and visualizes AI check-ins. It would be beneficial to observe how users interact with the interface in real time and see where the difficulties are at. The goal of *AdopTree* is to encourage a meaningful and consistent engagement from the user. Therefore, it is important for the user to understand the functionalities and see if they actually feel motivated and emotionally connected while interacting with their tree. The usability test would be made in person, where the user would be encouraged to think out loud. I would take notes on the specific difficulties that the users are facing. Each test would begin with a short introduction to the app's intentions, and then be followed with a set of predefined tasks. These tasks should reflect the most important things that the user should be able to do on the site. The tasks could be something like:

- Finding and selecting a tree near you on the tree map (pretend the map location works)
- Adopting a tree and see its profile
- Navigating to the "*My Trees*" page
- Doing a check-in on your tree by submitting a photo
- Understanding the AI health feedback and the tools to help
- Navigating to the *"Our Impact"* page

DISCUSSION

During the process of designing *AdopTree*, I learned a lot about the positives that comes from combining awareness about urban green spaces, with technology. One of the most important things I learned, was that people are often willing to help. They just don't always know how to do it or see the importance of their input. This was pointed out in the interviews and made it clear that my design should involve simplicity, accessibility, and personal/emotional motivation. On the other hand, there were also limitations to this design process. The interviews were relatively short and only included three participants, which meant I got limited response. Therefore, I missed perspectives from other types of users (e.g., elderly citizens), which could have been a very useful for my design. Another limitation was that I only have my first iteration of the high-fidelity prototype.

Even though I have implemented the initial requirements I observed in the interviews and from the literature, I don't have any feedback on the prototype, and this would be implemented in the future by doing a usability test.

CONCLUSIONS

During this design project I aimed to answer the question: *"How can you encourage people to engage in the care of urban trees, in an accessible and motivating way through technology?"*.

This answer was somewhat addressed through the final concept and the high-fidelity prototype, *AdopTree*. It combines accountability through tree adoption, accessibility with a location-based map and gives the user a step-by-step guide and the tools to care for their tree through the AI check-ins.

If I continued developing on *AdopTree*, the next step in my process would be a usability test of the prototype. This would provide information, about how intuitive and motivating the prototype is, to the different types of users. These insights could improve the interaction flow and possibly make it easier for users to understand how to help. This could include future features like reminders or a health history of the trees.

If I had to make the prototype into a real app, there would have to be some sort of collaboration with people who work with trees. This could be urban tree experts or environmental groups. To make sure that the AI features, like the instructions and the tools given are accurate, so the user actually helps the trees. The AI scanner would also need more development. It would have to be trained to recognize the signs of distress and diseases, for the different species of trees. One of the features of the prototype, is that the user can see the preexisting status of the urban trees near them. For that to actually work, all of the trees would have to be scanned and put into a database and this would require a lot of people and/or time or the feature needs to be changed.

Overall *AdopTree* is not just made to be a tree care app, it is also a platform for awareness and action. With further development, the concept has the potential to encourage people to care for trees and take action.

REFERENCES

[1] Affinity Diagramming: Collaboratively Sort UX Findings & Design Ideas: *https://www.nngroup.com/articles/affinity-diagram/*. Accessed: 2025-06-02.

[2] Ahlstrand, N.I., Larsen, J.C., Lillemark, M.R. and P. Tøttrup, A. 2022. Denmark explores: engaging citizen scientists nation-wide to monitor phenology. (Dec. 2022).

[3] Bytræer er en krævende opgave: 2013. *https://www.dr.dk/nyheder/viden/naturvidenskab/bytraeer-er-en-kraevende-opgave*. Accessed: 2025-06-01.

[4] Bytræer er med til at afbøde virkningerne af klimaændringer: 2009. *https://videntjenesten.ku.dk/park_og_landskab/planter/traeer/videnblad_08.01-22/*. Accessed: 2025-06-01.

[5] Farven grøn | Farvernes Betydning: *https://www.farvernesbetydning.dk/farven-gron/*. Accessed: 2025-06-02.

[6] Figma: The Collaborative Interface Design Tool: *https://www.figma.com/*. Accessed: 2025-06-01.

[7] Five ways to increase tree cover in cities (commentary): 2023. *https://news.mongabay.com/2023/07/five-ways-to-increase-tree-cover-in-cities-commentary/*. Accessed: 2025-06-01.

[8] Healthy Trees, Healthy Cities: *https://hthc.itreetools.org/home*. Accessed: 2025-06-01.

[9] High-Fidelity Prototyping: What Is It And How Can It Help? *https://www.figma.com/resource-library/high-fidelity-prototyping/*. Accessed: 2025-06-02.

[10] Home: 2023. *https://www.lincolninst.edu/publications/articles/2023-08-city-tech-tree-watering-apps-urban-forests-berlin-athens-dc-new-york/*. Accessed: 2025-06-03.

[11] Home: *https://plantnet.org/en/*. Accessed: 2025-06-01.

[12] How To Get a Prototype Made & Protect Your Design (2025): *https://www.shopify.com/blog/7076768-tim-ferriss-on-manufacturing-and-prototyping-a-product*. Accessed: 2025-06-02.

[13] Ltd, N.P. 2021. The Problems Urban Street Tree Face. *Citygreen*.

[14] Miro | Arbejdsområdet til innovation: *https://miro.com/da/*. Accessed: 2025-06-01.

[15] Task Scenarios for Usability Testing: *https://www.nngroup.com/articles/task-scenarios-usability-testing/*. Accessed: 2025-06-03.

[16] The 5 Stages in the Design Thinking Process: 2025. *https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process*. Accessed: 2025-06-01.

[17] User Requirements: *https://perfval.com/technical-discussion/user-requirements/*. Accessed: 2025-06-01.

[18] What is an Unstructured Interview? 2022. *https://delvetool.com/blog/unstructured*. Accessed: 2025-06-02.

[19] Why do we need trees? 2017. *https://www.woodlandtrust.org.uk/blog/2017/12/why-do-we-need-trees/*. Accessed: 2025-06-01.

[20] 2021. Exploratory Sketching. *Puck Verbeek*.

[21] Healthy Trees, Healthy Cities App. *Cities4Forests*.

[22] Read more. *Pl@ntNet*.